

Supplementary Theory and Algorithm Details for **exdqlm**

Antonio Aguirre

University of California Santa Cruz

Raquel Barata

University of California Santa Cruz

Raquel Prado

University of California Santa Cruz

Bruno Sansó

University of California Santa Cruz

Abstract

This supplement gives the mathematical details behind the inference algorithms implemented in **exdqlm**. It records model hierarchies, augmented posterior targets, Markov chain Monte Carlo (MCMC) updates, variational Bayes (VB) factorizations, and evidence lower bound (ELBO) components for dynamic quantile linear models (DQLMs) under asymmetric Laplace (AL) and extended asymmetric Laplace (exAL) likelihoods, and for static AL/exAL regression with ridge or regularized horseshoe normal-scale (RHS-NS) priors. The notation is aligned with the main article and with the package interfaces: vectors are written in bold lowercase, matrices in bold uppercase, and dimensions are stated when objects are introduced.

Keywords: Bayesian quantile regression, dynamic linear models, asymmetric Laplace likelihood, extended asymmetric Laplace likelihood, MCMC, variational Bayes.

1. Scope and organization

This supplement is a mathematical companion to the main article. It is intended for readers who want to inspect the posterior targets and update equations behind the package routines, rather than for users who only need the applied workflow shown in the examples. The dynamic sections cover AL-based DQLMs and exAL-based exDQLMs. The static sections cover AL/exAL regression with Gaussian ridge priors and with the RHS-NS sparse prior used by the package.

Each model section follows the same organization: model hierarchy, augmented posterior target, MCMC updates, VB factorization and coordinate updates, ELBO components, and the corresponding package function. The final section maps the main user-facing functions to these algorithmic blocks.

2. Notation and distributional building blocks

Let $p_0 \in (0, 1)$ be the target quantile. The response is scalar. In dynamic models, observations are indexed by $t = 1, \dots, T$; in static regression, observations are indexed by $i = 1, \dots, n$.

Lowercase bold symbols denote vectors, for example $\boldsymbol{\theta}_t \in \mathbb{R}^q$ and $\boldsymbol{\beta} \in \mathbb{R}^p$. Uppercase bold symbols denote matrices, for example $\mathbf{G}_t \in \mathbb{R}^{q \times q}$ and $\mathbf{X} \in \mathbb{R}^{n \times p}$. For any positive definite matrix $\mathbf{A}$, $|\mathbf{A}|$ denotes its determinant.

2.1. Notation contract

The supplement uses the following conventions throughout. The symbols p_γ , A_γ , B_γ , and C_γ are deterministic functions of (p_0, γ) , whereas $\pi_\gamma(\gamma)$ denotes the prior density for γ on (L, U) . Generic densities are denoted by $p(\cdot)$ only when no confusion with the exAL quantile map can arise. In variational sections, expectations without an explicit subscript are taken under the current mean-field distribution q . The symbol ξ is reserved for the RHS-NS half-Cauchy auxiliary variable; the transformed coordinate for γ in Laplace-Delta calculations is denoted by z_γ .

Symbol	Meaning
p_0	Target quantile level.
p_γ	exAL internal quantile map determined by (p_0, γ) .
A_0, B_0	AL constants $A(p_0)$ and $B(p_0)$.
$A_\gamma, B_\gamma, C_\gamma$	exAL constants evaluated at p_γ .
$\pi_\gamma(\gamma)$	Prior density for γ on (L, U) .
$\boldsymbol{\theta}_t, \mathbf{f}_t, \mathbf{G}_t, \mathbf{W}_t$	Dynamic state, observation vector, evolution matrix, and evolution covariance.
$\boldsymbol{\beta}, \mathbf{X}$	Static regression coefficients and design matrix.
$\mathcal{A}$	Active coefficient set receiving RHS-NS shrinkage.

Table S1: Core notation used throughout the supplement.

The distributional parameterizations are also fixed throughout: $\mathcal{N}(m, V)$ uses variance V ; $\text{IG}(a, b)$ uses rate b with density proportional to $x^{-(a+1)} \exp(-b/x)$; $\text{Exp}(\text{rate} = 1/\sigma)$ has density $\sigma^{-1} \exp(-x/\sigma)$; and $\mathcal{N}^+(m, V)$ denotes $\mathcal{N}(m, V)$ truncated to $(0, \infty)$. The generalized inverse Gaussian parameterization is given in (S8).

The asymmetric Laplace (AL) mixture representation used by the package is

$$v \mid \sigma \sim \text{Exp}(\text{rate} = 1/\sigma), \quad v > 0, \quad (\text{S1})$$

$$y \mid \eta, \sigma, v \sim \mathcal{N}\{\eta + A(p_0)v, \sigma B(p_0)v\}, \quad (\text{S2})$$

where

$$A(p) = \frac{1 - 2p}{p(1 - p)}, \quad B(p) = \frac{2}{p(1 - p)}. \quad (\text{S3})$$

The AL model is the $\gamma = 0$ special case of the exAL formulation below.

The exAL distribution is represented through an additional latent variable $s \sim \mathcal{N}^+(0, 1)$, a standard normal truncated to $(0, \infty)$. For $\gamma \in (L, U)$, define

$$g(\gamma) = 2\Phi(-|\gamma|) \exp(\gamma^2/2), \quad (\text{S4})$$

$$p_\gamma = \mathbb{I}(\gamma < 0) + \frac{p_0 - \mathbb{I}(\gamma < 0)}{g(\gamma)}, \quad (\text{S5})$$

$$A_\gamma = A(p_\gamma), \quad B_\gamma = B(p_\gamma), \quad C_\gamma = \{\mathbb{I}(\gamma > 0) - p_\gamma\}^{-1}. \quad (\text{S6})$$

The endpoints $L < 0 < U$ are the roots $g(L) = 1 - p_0$ and $g(U) = p_0$. Conditional on (v, s, σ, γ) ,

$$y \mid \eta, \sigma, \gamma, v, s \sim \mathcal{N}(\eta + C_\gamma \sigma |\gamma| s + A_\gamma v, \sigma B_\gamma v). \quad (\text{S7})$$

The package uses an inverse-gamma prior $\sigma \sim \text{IG}(a_\sigma, b_\sigma)$ and a proper truncated Student- t prior $\pi_\gamma(\gamma)$ for γ on (L, U) .

The generalized inverse Gaussian distribution is parameterized as

$$x \sim \text{GIG}(\lambda, \chi, \psi) \iff p(x) \propto x^{\lambda-1} \exp \left\{ -\frac{1}{2} \left(\frac{\chi}{x} + \psi x \right) \right\}, \quad x > 0. \quad (\text{S8})$$

For $r \in \mathbb{R}$,

$$\mathbb{E}(x^r) = \left(\frac{\chi}{\psi} \right)^{r/2} \frac{K_{\lambda+r}(\sqrt{\chi\psi})}{K_\lambda(\sqrt{\chi\psi})}, \quad (\text{S9})$$

where $K_\nu(\cdot)$ is the modified Bessel function of the second kind.

3. Dynamic DQLM under the AL likelihood

3.1. Model hierarchy and full joint

Let $\boldsymbol{\theta}_t \in \mathbb{R}^q$ be the latent state, $\mathbf{f}_t \in \mathbb{R}^q$ the observation vector, $\mathbf{G}_t \in \mathbb{R}^{q \times q}$ the evolution matrix, and $\mathbf{W}_t \in \mathbb{R}^{q \times q}$ a positive semidefinite evolution covariance. The dynamic DQLM is

$$\boldsymbol{\theta}_0 \sim \mathcal{N}(\mathbf{m}_0, \mathbf{C}_0), \quad \mathbf{m}_0 \in \mathbb{R}^q, \mathbf{C}_0 \in \mathbb{R}^{q \times q}, \quad (\text{S10})$$

$$\boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1} \sim \mathcal{N}(\mathbf{G}_t \boldsymbol{\theta}_{t-1}, \mathbf{W}_t), \quad (\text{S11})$$

$$v_t \mid \sigma \sim \text{Exp}(\text{rate} = 1/\sigma), \quad (\text{S12})$$

$$y_t \mid \boldsymbol{\theta}_t, \sigma, v_t \sim \mathcal{N}(\mathbf{f}_t^\top \boldsymbol{\theta}_t + A_0 v_t, \sigma B_0 v_t), \quad (\text{S13})$$

where $A_0 = A(p_0)$ and $B_0 = B(p_0)$. With $\mathbf{v} = (v_1, \dots, v_T)^\top$ and $\boldsymbol{\theta}_{0:T} = (\boldsymbol{\theta}_0, \dots, \boldsymbol{\theta}_T)$, the posterior target is

$$p(\boldsymbol{\theta}_{0:T}, \sigma, \mathbf{v} \mid \mathbf{y}) \propto p(\sigma) p(\boldsymbol{\theta}_0) \prod_{t=1}^T p(\boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1}) \prod_{t=1}^T p(v_t \mid \sigma) \prod_{t=1}^T p(y_t \mid \boldsymbol{\theta}_t, \sigma, v_t). \quad (\text{S14})$$

Equivalently, conditional on $(\sigma, \mathbf{v})$, the pseudo-observation $\tilde{y}_t = y_t - A_0 v_t$ satisfies

$$\tilde{y}_t = \mathbf{f}_t^\top \boldsymbol{\theta}_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma B_0 v_t). \quad (\text{S15})$$

Thus the conditional state posterior is a Gaussian state-space posterior.

3.2. MCMC updates

One Gibbs sweep for the dynamic DQLM uses the following full conditionals.

State path. Given $(\sigma, \mathbf{v})$, sample the state path $\boldsymbol{\theta}_{0:T}$ exactly by applying forward-filtering backward-sampling (FFBS) to (S15).

Scale. Let $r_t = y_t - \mathbf{f}_t^\top \boldsymbol{\theta}_t$. Then

$$\sigma \mid \boldsymbol{\theta}_{0:T}, \mathbf{v}, \mathbf{y} \sim \text{IG} \left(a_\sigma + \frac{3T}{2}, b_\sigma + \sum_{t=1}^T v_t + \sum_{t=1}^T \frac{(r_t - A_0 v_t)^2}{2B_0 v_t} \right). \quad (\text{S16})$$

Latent AL variables. For $t = 1, \dots, T$,

$$v_t \mid \boldsymbol{\theta}_t, \sigma, y_t \sim \text{GIG} \left(\frac{1}{2}, \frac{r_t^2}{\sigma B_0}, \frac{A_0^2}{\sigma B_0} + \frac{2}{\sigma} \right). \quad (\text{S17})$$

3.3. Mean-field VB and ELBO

The dynamic DQLM VB approximation uses

$$q(\boldsymbol{\theta}_{0:T}, \sigma, \mathbf{v}) = q(\boldsymbol{\theta}_{0:T}) q(\sigma) \prod_{t=1}^T q(v_t), \quad (\text{S18})$$

with $q(\sigma) = \text{IG}(a_q, b_q)$ and $q(v_t) = \text{GIG}(1/2, \chi_t, \psi_t)$. Define

$$\kappa = \mathbb{E}_q(\sigma^{-1}) = \frac{a_q}{b_q}, \quad \ell_t = \mathbb{E}_q(v_t^{-1}), \quad \nu_t = \mathbb{E}_q(v_t). \quad (\text{S19})$$

The state factor is the Gaussian posterior of the auxiliary model

$$\mathbf{y}_t^* = \mathbf{f}_t^\top \boldsymbol{\theta}_t + \epsilon_t^*, \quad \epsilon_t^* \sim \mathcal{N}(0, R_t^*), \quad \mathbf{y}_t^* = y_t - \frac{A_0}{\ell_t}, \quad R_t^* = \frac{B_0}{\kappa \ell_t}. \quad (\text{S20})$$

After Kalman smoothing, write $q(\boldsymbol{\theta}_t) = \mathcal{N}(\mathbf{m}_{t|T}, \mathbf{C}_{t|T})$ and define

$$\bar{r}_t = y_t - \mathbf{f}_t^\top \mathbf{m}_{t|T}, \quad (\text{S21})$$

$$\bar{r}_t^2 = \bar{r}_t^2 + \mathbf{f}_t^\top \mathbf{C}_{t|T} \mathbf{f}_t. \quad (\text{S22})$$

The remaining coordinate-ascent variational inference (CAVI) updates are

$$q(v_t) = \text{GIG} \left(\frac{1}{2}, \frac{\kappa}{B_0} \bar{r}_t^2, \kappa \left(2 + \frac{A_0^2}{B_0} \right) \right), \quad (\text{S23})$$

$$q(\sigma) = \text{IG}(a_q, b_q), \quad a_q = a_\sigma + \frac{3T}{2}, \quad (\text{S24})$$

$$b_q = b_\sigma + \sum_{t=1}^T \nu_t + \frac{1}{2B_0} \sum_{t=1}^T \left\{ \ell_t \bar{r}_t^2 - 2A_0 \bar{r}_t + A_0^2 \nu_t \right\}. \quad (\text{S25})$$

The ELBO is

$$\mathcal{L}(q) = \mathbb{E}_q \{ \log p(\mathbf{y}, \boldsymbol{\theta}_{0:T}, \sigma, \mathbf{v}) \} - \mathbb{E}_q \{ \log q(\boldsymbol{\theta}_{0:T}, \sigma, \mathbf{v}) \} \quad (\text{S26})$$

$$= \mathcal{L}_\theta + \mathbb{E}_q \{ \log p(\sigma) \} + \sum_{t=1}^T \mathbb{E}_q \{ \log p(v_t \mid \sigma) \} + \sum_{t=1}^T \mathbb{E}_q \{ \log p(y_t \mid \boldsymbol{\theta}_t, \sigma, v_t) \} \quad (\text{S27})$$

$$+ H \{ q(\sigma) \} + \sum_{t=1}^T H \{ q(v_t) \}. \quad (\text{S28})$$

The state-path term $\mathcal{L}_\theta$ is evaluated from the auxiliary Gaussian state-space model in (S20) through the Kalman marginal likelihood identity used in the package ELBO diagnostics.

4. Dynamic exDQLM under the exAL likelihood

4.1. Model hierarchy and full joint

The dynamic exDQLM replaces the AL observation in (S13) with the exAL augmentation

$$v_t \mid \sigma \sim \text{Exp}(\text{rate} = 1/\sigma), \quad s_t \sim \mathcal{N}^+(0, 1), \quad (\text{S29})$$

$$y_t \mid \boldsymbol{\theta}_t, \sigma, \gamma, v_t, s_t \sim \mathcal{N}\left(\mathbf{f}_t^\top \boldsymbol{\theta}_t + C_\gamma \sigma |\gamma| s_t + A_\gamma v_t, \sigma B_\gamma v_t\right), \quad (\text{S30})$$

with the same state evolution as in Section 3. The complete posterior target is

$$\begin{aligned} p(\boldsymbol{\theta}_{0:T}, \mathbf{v}, \mathbf{s}, \sigma, \gamma \mid \mathbf{y}) &\propto p(\boldsymbol{\theta}_0) \prod_{t=1}^T p(\boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1}) \prod_{t=1}^T p(y_t \mid \boldsymbol{\theta}_t, v_t, s_t, \sigma, \gamma) \\ &\times \prod_{t=1}^T p(v_t \mid \sigma) \prod_{t=1}^T p(s_t) p(\sigma) \pi_\gamma(\gamma). \end{aligned} \quad (\text{S31})$$

Conditional on $(\mathbf{v}, \mathbf{s}, \sigma, \gamma)$,

$$\tilde{y}_t = y_t - A_\gamma v_t - C_\gamma \sigma |\gamma| s_t, \quad R_t = \sigma B_\gamma v_t, \quad (\text{S32})$$

which again gives a linear Gaussian state-space model for $\boldsymbol{\theta}_{0:T}$.

4.2. MCMC updates

The default package path samples σ from its exact conditional given γ and then updates γ by bounded univariate slice sampling. Joint random-walk alternatives are also implemented, but the following target is the default exact-kernel path.

Latent v_t . Let $r_t = y_t - \mathbf{f}_t^\top \boldsymbol{\theta}_t - C_\gamma \sigma |\gamma| s_t$. Then

$$v_t \mid \cdot \sim \text{GIG}\left(\frac{1}{2}, \frac{r_t^2}{\sigma B_\gamma}, \frac{A_\gamma^2}{\sigma B_\gamma} + \frac{2}{\sigma}\right). \quad (\text{S33})$$

Latent s_t . Let $y_t^\circ = y_t - \mathbf{f}_t^\top \boldsymbol{\theta}_t - A_\gamma v_t$ and $d_\gamma = C_\gamma \sigma |\gamma|$. Then

$$s_t \mid \cdot \sim \mathcal{N}^+(m_{s,t}, V_{s,t}), \quad V_{s,t} = \left(1 + \frac{d_\gamma^2}{\sigma B_\gamma v_t}\right)^{-1}, \quad m_{s,t} = V_{s,t} \frac{d_\gamma y_t^\circ}{\sigma B_\gamma v_t}. \quad (\text{S34})$$

State path. Given $(\mathbf{v}, \mathbf{s}, \sigma, \gamma)$, sample $\boldsymbol{\theta}_{0:T}$ by FFBS using (S32).

Scale. For fixed γ , let $y_t^\circ = y_t - \mathbf{f}_t^\top \boldsymbol{\theta}_t - A_\gamma v_t$ and $u_t = C_\gamma |\gamma| s_t$. Then

$$\sigma \mid \cdot \sim \text{GIG}(\lambda_\sigma, \chi_\sigma, \psi_\sigma), \quad (\text{S35})$$

where

$$\lambda_\sigma = - \left(a_\sigma + \frac{3T}{2} \right), \quad (\text{S36})$$

$$\chi_\sigma = 2b_\sigma + 2 \sum_{t=1}^T v_t + \sum_{t=1}^T \frac{(y_t^\circ)^2}{B_\gamma v_t}, \quad (\text{S37})$$

$$\psi_\sigma = \sum_{t=1}^T \frac{u_t^2}{B_\gamma v_t}. \quad (\text{S38})$$

Skewness. The conditional kernel for γ is

$$\pi(\gamma \mid \cdot) \propto \pi_\gamma(\gamma) \prod_{t=1}^T (\sigma B_\gamma v_t)^{-1/2} \exp \left[- \sum_{t=1}^T \frac{\{y_t - \mathbf{f}_t^\top \boldsymbol{\theta}_t - A_\gamma v_t - C_\gamma \sigma |\gamma| s_t\}^2}{2\sigma B_\gamma v_t} \right], \quad (\text{S39})$$

for $\gamma \in (L, U)$. The default sampler applies the bounded slice update in Section 7 directly to this kernel.

4.3. Laplace-Delta VB approximation

The mean-field family is

$$q(\boldsymbol{\theta}_{0:T}, \mathbf{v}, \mathbf{s}, \sigma, \gamma) = q(\boldsymbol{\theta}_{0:T}) \prod_{t=1}^T q(v_t) \prod_{t=1}^T q(s_t) q(\sigma, \gamma). \quad (\text{S40})$$

The conjugate factors are updated conditionally on moments from $q(\sigma, \gamma)$. Define

$$\kappa_1 = \mathbb{E}\{1/(\sigma B_\gamma)\}, \quad \kappa_2 = \mathbb{E}\{A_\gamma/(\sigma B_\gamma)\}, \quad \kappa_3 = \mathbb{E}\{A_\gamma^2/(\sigma B_\gamma)\}, \quad (\text{S41})$$

$$\kappa_4 = \mathbb{E}\{C_\gamma |\gamma|/B_\gamma\}, \quad \kappa_5 = \mathbb{E}\{\sigma C_\gamma^2 \gamma^2/B_\gamma\}, \quad \kappa_6 = \mathbb{E}\{A_\gamma C_\gamma |\gamma|/B_\gamma\}. \quad (\text{S42})$$

Let $m_{1/v,t} = \mathbb{E}(v_t^{-1})$, $m_{s,t} = \mathbb{E}(s_t)$, and $m_{s^2,t} = \mathbb{E}(s_t^2)$. The state factor is Gaussian and is computed by Kalman smoothing with observation precision

$$\phi_t = \kappa_1 m_{1/v,t} \quad (\text{S43})$$

and pseudo-response

$$y_t^{\text{VB}} = \frac{y_t \phi_t - \kappa_2 - \kappa_4 m_{s,t} m_{1/v,t}}{\phi_t}. \quad (\text{S44})$$

If $\eta_t = \mathbf{f}_t^\top \boldsymbol{\theta}_t$, write $m_{\eta,t} = \mathbb{E}(\eta_t)$, $S_{\eta,t} = \mathbb{V}(\eta_t)$, $m_{\Delta,t} = y_t - m_{\eta,t}$, and $S_{\Delta,t} = m_{\Delta,t}^2 + S_{\eta,t}$. Then

$$q(v_t) = \text{GIG} \left(\frac{1}{2}, \kappa_1 S_{\Delta,t} - 2\kappa_4 m_{\Delta,t} m_{s,t} + \kappa_5 m_{s^2,t}, \kappa_3 + 2\mathbb{E}(\sigma^{-1}) \right), \quad (\text{S45})$$

$$q(s_t) = \mathcal{N}^+(m_{s,t}^{\text{VB}}, V_{s,t}^{\text{VB}}), \quad (\text{S46})$$

$$V_{s,t}^{\text{VB}} = \left(1 + \kappa_5 m_{1/v,t} \right)^{-1}, \quad (\text{S47})$$

$$m_{s,t}^{\text{VB}} = V_{s,t}^{\text{VB}} \left(\kappa_4 m_{\Delta,t} m_{1/v,t} - \kappa_6 \right). \quad (\text{S48})$$

The nonconjugate factor $q(\sigma, \gamma)$ is approximated by a Laplace-Delta step on transformed coordinates

$$u = \log \sigma, \quad z_\gamma = \text{logit} \left(\frac{\gamma - L}{U - L} \right). \quad (\text{S49})$$

The transformed objective is the expected complete log-joint plus the Jacobian

$$u + \log \{d\gamma/dz_\gamma\}. \quad (\text{S50})$$

The package stores the transformed mode, the approximate covariance matrix, Delta-method moments, and ELBO components.

ELBO. The LDVB ELBO has the block decomposition

$$\mathcal{L}(q) = \mathcal{L}_{\text{like}} + \mathcal{L}_\theta + \mathcal{L}_v + \mathcal{L}_s + \mathcal{L}_{\sigma\gamma} + H\{q(\boldsymbol{\theta}_{0:T})\} + \sum_{t=1}^T H\{q(v_t)\} + \sum_{t=1}^T H\{q(s_t)\} + H\{q(\sigma, \gamma)\}. \quad (\text{S51})$$

The state and latent-variable blocks are exact for the current $q(\sigma, \gamma)$ moments; the (σ, γ) entropy and smooth expectations are evaluated under the Laplace-Delta approximation.

5. Static AL and exAL regression with ridge priors

5.1. Model hierarchy and full joint

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ have row vectors $\mathbf{x}_i^\top$, and let $\boldsymbol{\beta} \in \mathbb{R}^p$. The ridge/Gaussian coefficient prior is

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{b}_0, \mathbf{B}_0), \quad \mathbf{b}_0 \in \mathbb{R}^p, \quad \mathbf{B}_0 \in \mathbb{R}^{p \times p}. \quad (\text{S52})$$

Under the exAL likelihood,

$$v_i \mid \sigma \sim \text{Exp}(\text{rate} = 1/\sigma), \quad s_i \sim \mathcal{N}^+(0, 1), \quad (\text{S53})$$

$$y_i \mid \boldsymbol{\beta}, \sigma, \gamma, v_i, s_i \sim \mathcal{N}(\mathbf{x}_i^\top \boldsymbol{\beta} + C_\gamma \sigma |\gamma| s_i + A_\gamma v_i, \sigma B_\gamma v_i). \quad (\text{S54})$$

The static AL model is obtained by setting $\gamma = 0$, removing s_i , and using A_0, B_0 . The exAL full joint is

$$p(\boldsymbol{\beta}, \sigma, \gamma, \mathbf{v}, \mathbf{s}, \mathbf{y}) = p(\boldsymbol{\beta})p(\sigma)\pi_\gamma(\gamma) \prod_{i=1}^n p(y_i \mid \boldsymbol{\beta}, \sigma, \gamma, v_i, s_i)p(v_i \mid \sigma)p(s_i). \quad (\text{S55})$$

5.2. MCMC updates

For fixed $(\sigma, \gamma, \mathbf{v}, \mathbf{s})$, define

$$y_i^* = y_i - C_\gamma \sigma |\gamma| s_i - A_\gamma v_i, \quad w_i = (\sigma B_\gamma v_i)^{-1}. \quad (\text{S56})$$

Let $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$ and $\mathbf{y}^* = (y_1^*, \dots, y_n^*)^\top$. Then

$$\boldsymbol{\Sigma}_{\boldsymbol{\beta}|\cdot} = (\mathbf{B}_0^{-1} + \mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}, \quad (\text{S57})$$

$$\mathbf{m}_{\boldsymbol{\beta}|\cdot} = \boldsymbol{\Sigma}_{\boldsymbol{\beta}|\cdot} (\mathbf{B}_0^{-1} \mathbf{b}_0 + \mathbf{X}^\top \mathbf{W} \mathbf{y}^*), \quad (\text{S58})$$

$$\boldsymbol{\beta} \mid \cdot \sim \mathcal{N}(\mathbf{m}_{\boldsymbol{\beta}|\cdot}, \boldsymbol{\Sigma}_{\boldsymbol{\beta}|\cdot}). \quad (\text{S59})$$

Let $D_\gamma = C_\gamma |\gamma|$. The remaining exAL full conditionals are

$$s_i \mid \cdot \sim \mathcal{N}^+(m_{s,i}, V_{s,i}), \quad (\text{S60})$$

$$V_{s,i} = \left(1 + \frac{\sigma D_\gamma^2}{B_\gamma v_i}\right)^{-1}, \quad (\text{S61})$$

$$m_{s,i} = V_{s,i} \frac{D_\gamma (y_i - \mathbf{x}_i^\top \boldsymbol{\beta} - A_\gamma v_i)}{B_\gamma v_i}, \quad (\text{S62})$$

and, with $r_i = y_i - \mathbf{x}_i^\top \boldsymbol{\beta} - \sigma D_\gamma s_i$,

$$v_i \mid \cdot \sim \text{GIG} \left(\frac{1}{2}, \frac{r_i^2}{\sigma B_\gamma}, \frac{A_\gamma^2}{\sigma B_\gamma} + \frac{2}{\sigma} \right). \quad (\text{S63})$$

For fixed γ , define $t_i = y_i - \mathbf{x}_i^\top \boldsymbol{\beta} - A_\gamma v_i$. Then

$$\sigma \mid \cdot \sim \text{GIG} \left(-a_\sigma - \frac{3n}{2}, 2b_\sigma + 2 \sum_{i=1}^n v_i + \sum_{i=1}^n \frac{t_i^2}{B_\gamma v_i}, \sum_{i=1}^n \frac{D_\gamma^2 s_i^2}{B_\gamma v_i} \right). \quad (\text{S64})$$

The nonstandard γ kernel is

$$\pi(\gamma \mid \cdot) \propto \pi_\gamma(\gamma) [B_\gamma]^{-n/2} \exp \left\{ \sum_{i=1}^n \frac{D_\gamma s_i t_i}{B_\gamma v_i} - \frac{1}{2\sigma} \sum_{i=1}^n \frac{t_i^2}{B_\gamma v_i} - \frac{\sigma}{2} \sum_{i=1}^n \frac{D_\gamma^2 s_i^2}{B_\gamma v_i} \right\}, \quad (\text{S65})$$

on (L, U) . The package default updates this one-dimensional target by the bounded slice sampler in Section 7.

For the AL special case, the Gibbs updates reduce to

$$\boldsymbol{\beta} \mid \cdot \sim \mathcal{N}(\mathbf{m}_{\boldsymbol{\beta}|\cdot}, \boldsymbol{\Sigma}_{\boldsymbol{\beta}|\cdot}), \quad (\text{S66})$$

$$\sigma \mid \cdot \sim \text{IG} \left(a_\sigma + \frac{3n}{2}, b_\sigma + \sum_{i=1}^n v_i + \sum_{i=1}^n \frac{(y_i - \mathbf{x}_i^\top \boldsymbol{\beta} - A_0 v_i)^2}{2B_0 v_i} \right), \quad (\text{S67})$$

$$v_i \mid \cdot \sim \text{GIG} \left(\frac{1}{2}, \frac{(y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2}{\sigma B_0}, \frac{A_0^2}{\sigma B_0} + \frac{2}{\sigma} \right). \quad (\text{S68})$$

5.3. Laplace-Delta VB updates and ELBO

The static exAL variational family is

$$q(\boldsymbol{\beta}, \mathbf{v}, \mathbf{s}, \sigma, \gamma) = q(\boldsymbol{\beta}) \prod_{i=1}^n q(v_i) \prod_{i=1}^n q(s_i) q(\sigma, \gamma). \quad (\text{S69})$$

The coefficient factor is Gaussian. With the same moment definitions as in (S42),

$$\omega_i = \kappa_1 \mathbb{E}(v_i^{-1}), \quad (\text{S70})$$

$$\eta_i = y_i \kappa_1 \mathbb{E}(v_i^{-1}) - \kappa_4 \mathbb{E}(v_i^{-1}) \mathbb{E}(s_i) - \kappa_2, \quad (\text{S71})$$

$$\boldsymbol{\Sigma}_\beta = \left(\mathbf{B}_0^{-1} + \mathbf{X}^\top \text{diag}(\omega_1, \dots, \omega_n) \mathbf{X} \right)^{-1}, \quad (\text{S72})$$

$$\mathbf{m}_\beta = \boldsymbol{\Sigma}_\beta \left(\mathbf{B}_0^{-1} \mathbf{b}_0 + \mathbf{X}^\top \boldsymbol{\eta} \right), \quad (\text{S73})$$

where $\boldsymbol{\eta} = (\eta_1, \dots, \eta_n)^\top$. The $q(v_i)$ and $q(s_i)$ updates are also closed form conditional on moments of $q(\sigma, \gamma)$. Let

$$\bar{r}_i = y_i - \mathbf{x}_i^\top \mathbf{m}_\beta, \quad \bar{r}_i^2 = r_i^2 + \mathbf{x}_i^\top \boldsymbol{\Sigma}_\beta \mathbf{x}_i. \quad (\text{S74})$$

Then

$$q(v_i) = \text{GIG} \left(\frac{1}{2}, \kappa_1 \bar{r}_i^2 - 2\kappa_4 \bar{r}_i \mathbb{E}(s_i) + \kappa_5 \mathbb{E}(s_i^2), \kappa_3 + 2\mathbb{E}(\sigma^{-1}) \right), \quad (\text{S75})$$

$$q(s_i) = \mathcal{N}^+(m_{s,i}^{\text{VB}}, V_{s,i}^{\text{VB}}), \quad (\text{S76})$$

$$V_{s,i}^{\text{VB}} = \left(1 + \kappa_5 \mathbb{E}(v_i^{-1}) \right)^{-1}, \quad (\text{S77})$$

$$m_{s,i}^{\text{VB}} = V_{s,i}^{\text{VB}} \left\{ \kappa_4 \bar{r}_i \mathbb{E}(v_i^{-1}) - \kappa_6 \right\}. \quad (\text{S78})$$

The nonconjugate factor $q(\sigma, \gamma)$ is proportional to

$$q(\sigma, \gamma) \propto \pi_\gamma(\gamma) [B_\gamma]^{-n/2} \sigma^{-(a_\sigma + 1 + 3n/2)} \exp \left\{ -\frac{1}{2} \left(\frac{\bar{c}(\gamma)}{\sigma} + \bar{d}(\gamma) \sigma \right) + \bar{e}(\gamma) \right\}, \quad (\text{S79})$$

where

$$\bar{c}(\gamma) = 2b_\sigma + 2 \sum_{i=1}^n \mathbb{E}(v_i) + \sum_{i=1}^n \frac{\mathbb{E}\{(y_i - \mathbf{x}_i^\top \boldsymbol{\beta} - A_\gamma v_i)^2 / v_i\}}{B_\gamma}, \quad (\text{S80})$$

$$\bar{d}(\gamma) = \sum_{i=1}^n \frac{C_\gamma^2 \gamma^2}{B_\gamma} \mathbb{E}\{s_i^2 / v_i\}, \quad (\text{S81})$$

$$\bar{e}(\gamma) = \sum_{i=1}^n \frac{C_\gamma |\gamma|}{B_\gamma} \mathbb{E}\{s_i (y_i - \mathbf{x}_i^\top \boldsymbol{\beta} - A_\gamma v_i) / v_i\}. \quad (\text{S82})$$

This factor is approximated by the same Laplace-Delta method as in the dynamic exDQLM. The AL path sets $\gamma = 0$ and uses $q(\boldsymbol{\beta})q(\sigma) \prod_i q(v_i)$ with the same closed-form updates as Section 3, replacing state smoothing moments by $\mathbb{E}\{(y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2\}$.

The static exAL ELBO is

$$\mathcal{L}(q) = \mathbb{E}_q \{ \log p(\boldsymbol{\beta}) + \log p(\sigma) + \log \pi_\gamma(\gamma) \} + \sum_{i=1}^n \mathbb{E}_q \{ \log p(y_i | \cdot) + \log p(v_i | \sigma) + \log p(s_i) \} + H(q), \quad (\text{S83})$$

where $H(q)$ is the sum of the Gaussian, GIG, truncated-normal, and Laplace-Delta entropy terms.

6. Static regression with the RHS-NS prior

This section focuses on the `rhs_ns` coefficient prior, the Gibbs-friendly regularized horseshoe variant used for sparse static regression. It enters the static Gaussian update for $\boldsymbol{\beta}$ through a prior precision matrix. Let $\mathcal{A} \subseteq \{1, \dots, p\}$ denote the active shrinkage set; by default an intercept, if present, is not shrunk. For $j \notin \mathcal{A}$ the package uses a fixed intercept precision. The direct `rhs` path is not developed here because its log-scale block is nonconjugate; the `rhs_ns` path is the conjugate sparse-prior formulation documented in this supplement.

6.1. RHS-NS prior hierarchy

For $j \in \mathcal{A}$, the `rhs_ns` hierarchy is

$$\beta_j \mid \tau^2, \lambda_j^2, \zeta^2 \propto \mathcal{N}(\beta_j \mid 0, \tau^2 \lambda_j^2) \mathcal{N}(\beta_j \mid 0, \zeta^2), \quad (\text{S84})$$

$$\lambda_j^2 \mid \nu_j \sim \text{IG}(1/2, 1/\nu_j), \quad \nu_j \sim \text{IG}(1/2, 1), \quad (\text{S85})$$

$$\tau^2 \mid \xi \sim \text{IG}(1/2, 1/\xi), \quad \xi \sim \text{IG}(1/2, 1/\tau_0^2), \quad (\text{S86})$$

$$\zeta^2 \sim \text{IG}(a_\zeta, b_\zeta), \quad (\text{S87})$$

unless ζ^2 is fixed by the user. The resulting prior precision contribution for active coefficients is

$$\pi_j = \frac{1}{\tau^2 \lambda_j^2} + \frac{1}{\zeta^2}. \quad (\text{S88})$$

Consequently, the β update in MCMC or VB is obtained from the static Gaussian update after replacing $\mathbf{B}_0^{-1}$ by a diagonal precision matrix with active entries π_j or their variational expectations.

6.2. RHS-NS MCMC updates

Conditioning on β , the shrinkage block has closed-form inverse-gamma updates:

$$\lambda_j^2 \mid \cdot \sim \text{IG}\left(1, \frac{1}{\nu_j} + \frac{\beta_j^2}{2\tau^2}\right), \quad (\text{S89})$$

$$\nu_j \mid \cdot \sim \text{IG}\left(1, 1 + \frac{1}{\lambda_j^2}\right), \quad (\text{S90})$$

$$\tau^2 \mid \cdot \sim \text{IG}\left(\frac{|\mathcal{A}|+1}{2}, \frac{1}{\xi} + \frac{1}{2} \sum_{j \in \mathcal{A}} \frac{\beta_j^2}{\lambda_j^2}\right), \quad (\text{S91})$$

$$\xi \mid \cdot \sim \text{IG}\left(1, \frac{1}{\tau_0^2} + \frac{1}{\tau^2}\right), \quad (\text{S92})$$

$$\zeta^2 \mid \cdot \sim \text{IG}\left(a_\zeta + \frac{|\mathcal{A}|}{2}, b_\zeta + \frac{1}{2} \sum_{j \in \mathcal{A}} \beta_j^2\right), \quad (\text{S93})$$

with the last line omitted when ζ^2 is fixed. These equations are the conditional targets for the RHS-NS scale block. The implementation may temporarily freeze or schedule the global-scale update during configured warmup iterations for numerical stability; this changes only the update schedule, not the conditional target displayed above.

6.3. RHS-NS VB updates and ELBO contribution

The package uses a mean-field approximation over the RHS-NS scale block. For $j \in \mathcal{A}$,

$$q(\lambda_j^2) = \text{IG}(a_{\lambda_j}, b_{\lambda_j}), \quad q(\nu_j) = \text{IG}(a_{\nu_j}, b_{\nu_j}), \quad (\text{S94})$$

$$q(\tau^2) = \text{IG}(a_\tau, b_\tau), \quad q(\xi) = \text{IG}(a_\xi, b_\xi), \quad (\text{S95})$$

$$q(\zeta^2) = \text{IG}(a_\zeta^*, b_\zeta^*), \quad (\text{S96})$$

with $a_{\lambda_j} = a_{\nu_j} = a_\xi = 1$ and $a_\tau = (|\mathcal{A}| + 1)/2$. Let $\bar{\beta}_j^2 = \mathbb{E}_q(\beta_j^2)$. The coordinate updates are

$$b_{\lambda_j} = \frac{1}{2} \bar{\beta}_j^2 \mathbb{E}(\tau^{-2}) + \mathbb{E}(\nu_j^{-1}), \quad (\text{S97})$$

$$b_{\nu_j} = 1 + \mathbb{E}\{(\lambda_j^2)^{-1}\}, \quad (\text{S98})$$

$$b_\tau = \frac{1}{2} \sum_{j \in \mathcal{A}} \bar{\beta}_j^2 \mathbb{E}\{(\lambda_j^2)^{-1}\} + \mathbb{E}(\xi^{-1}), \quad (\text{S99})$$

$$b_\xi = \frac{1}{\tau_0^2} + \mathbb{E}(\tau^{-2}), \quad (\text{S100})$$

$$a_\zeta^* = a_\zeta + \frac{|\mathcal{A}|}{2}, \quad b_\zeta^* = b_\zeta + \frac{1}{2} \sum_{j \in \mathcal{A}} \bar{\beta}_j^2. \quad (\text{S101})$$

The third line is shorthand for $b_\tau = \frac{1}{2} \sum_{j \in \mathcal{A}} \bar{\beta}_j^2 \mathbb{E}\{(\lambda_j^2)^{-1}\} + \mathbb{E}(\xi^{-1})$, where the expectation of $(\lambda_j^2)^{-1}$ is component-specific inside the summation. As in MCMC, optional warmup scheduling for the τ^2, ξ block affects when the coordinate is updated, not the coordinate target itself. The following term-by-term ELBO contribution matches the package's `rhs_ns` scale block. For $x \sim \text{IG}(a, b)$ under the rate parameterization used above, define

$$m_{\log x}(a, b) = \log b - \psi(a), \quad m_{x^{-1}}(a, b) = a/b, \quad (\text{S102})$$

$$h_{\text{IG}}(a, b) = a + \log b + \log \Gamma(a) - (a + 1)\psi(a), \quad (\text{S103})$$

where $\psi(\cdot)$ is the digamma function and h_{IG} is the entropy of an inverse-gamma variational factor. Let

$$\ell_{\lambda_j} = m_{\log x}(a_{\lambda_j}, b_{\lambda_j}), \quad r_{\lambda_j} = m_{x^{-1}}(a_{\lambda_j}, b_{\lambda_j}), \quad (\text{S104})$$

$$\ell_{\nu_j} = m_{\log x}(a_{\nu_j}, b_{\nu_j}), \quad r_{\nu_j} = m_{x^{-1}}(a_{\nu_j}, b_{\nu_j}), \quad (\text{S105})$$

$$\ell_\tau = m_{\log x}(a_\tau, b_\tau), \quad r_\tau = m_{x^{-1}}(a_\tau, b_\tau), \quad (\text{S106})$$

$$\ell_\xi = m_{\log x}(a_\xi, b_\xi), \quad r_\xi = m_{x^{-1}}(a_\xi, b_\xi). \quad (\text{S107})$$

If ζ^2 is random, set $\ell_\zeta = m_{\log x}(a_\zeta^*, b_\zeta^*)$ and $r_\zeta = m_{x^{-1}}(a_\zeta^*, b_\zeta^*)$; if it is fixed at ζ_0^2 , set $\ell_\zeta = \log \zeta_0^2$ and $r_\zeta = 1/\zeta_0^2$.

The coefficient-prior contribution is

$$\mathcal{L}_{\beta, \text{hs}} = \sum_{j \in \mathcal{A}} \left\{ -\frac{1}{2} \log(2\pi) - \frac{1}{2} \ell_\tau - \frac{1}{2} \ell_{\lambda_j} - \frac{1}{2} \bar{\beta}_j^2 r_\tau r_{\lambda_j} \right\}, \quad (\text{S108})$$

$$\mathcal{L}_{\beta, \text{slab}} = \sum_{j \in \mathcal{A}} \left\{ -\frac{1}{2} \log(2\pi) - \frac{1}{2} \ell_\zeta - \frac{1}{2} \bar{\beta}_j^2 r_\zeta \right\}. \quad (\text{S109})$$

The local, global, and slab prior contributions are

$$\mathcal{L}_{\lambda|\nu} = \sum_{j \in \mathcal{A}} \left\{ -\frac{1}{2} \ell_{\nu j} - \log \Gamma(1/2) - \frac{3}{2} \ell_{\lambda j} - r_{\nu j} r_{\lambda j} \right\}, \quad (\text{S110})$$

$$\mathcal{L}_{\nu} = \sum_{j \in \mathcal{A}} \left\{ -\log \Gamma(1/2) - \frac{3}{2} \ell_{\nu j} - r_{\nu j} \right\}, \quad (\text{S111})$$

$$\mathcal{L}_{\tau|\xi} = -\frac{1}{2} \ell_{\xi} - \log \Gamma(1/2) - \frac{3}{2} \ell_{\tau} - r_{\xi} r_{\tau}, \quad (\text{S112})$$

$$\mathcal{L}_{\xi} = \frac{1}{2} \log(\tau_0^{-2}) - \log \Gamma(1/2) - \frac{3}{2} \ell_{\xi} - \tau_0^{-2} r_{\xi}, \quad (\text{S113})$$

$$\mathcal{L}_{\zeta} = \begin{cases} a_{\zeta} \log b_{\zeta} - \log \Gamma(a_{\zeta}) - (a_{\zeta} + 1) \ell_{\zeta} - b_{\zeta} r_{\zeta}, & \zeta^2 \text{ random,} \\ 0, & \zeta^2 \text{ fixed.} \end{cases} \quad (\text{S114})$$

The entropy contribution is

$$\mathcal{L}_{\text{ent}} = \sum_{j \in \mathcal{A}} h_{\text{IG}}(a_{\lambda j}, b_{\lambda j}) + \sum_{j \in \mathcal{A}} h_{\text{IG}}(a_{\nu j}, b_{\nu j}) + h_{\text{IG}}(a_{\tau}, b_{\tau}) + h_{\text{IG}}(a_{\xi}, b_{\xi}) + \mathbb{I}(\zeta^2 \text{ random}) h_{\text{IG}}(a_{\zeta}^*, b_{\zeta}^*). \quad (\text{S115})$$

Finally, when the intercept is not included in $\mathcal{A}$, the package adds the fixed Gaussian intercept-prior contribution

$$\mathcal{L}_{\beta_0} = \frac{1}{2} \{ \log(\pi_0) - \log(2\pi) \} - \frac{1}{2} \pi_0 \bar{\beta}_0^2, \quad (\text{S116})$$

where π_0 is the fixed intercept precision. If the intercept is shrunk, this term is omitted because the intercept is already represented in the active RHS-NS block.

Thus the package contribution is

$$\mathcal{L}_{\text{rhs_ns}} = \mathcal{L}_{\beta, \text{hs}} + \mathcal{L}_{\beta, \text{slab}} + \mathcal{L}_{\lambda|\nu} + \mathcal{L}_{\nu} + \mathcal{L}_{\tau|\xi} + \mathcal{L}_{\xi} + \mathcal{L}_{\zeta} + \mathcal{L}_{\text{ent}} + \mathcal{L}_{\beta_0}. \quad (\text{S117})$$

All terms are available in closed form because each scale factor is inverse-gamma. This is the RHS-NS component of the full static-model ELBO; the likelihood, latent-variable, and (σ, γ) terms are added by the surrounding static AL or exAL variational routine.

7. Bounded slice sampling and Laplace-Delta blocks

7.1. Bounded univariate slice sampler

Let $h(x)$ be an unnormalized log-density on a finite interval $[a, b]$, and let $x_0 \in (a, b)$. The bounded slice update implemented in the package is:

- (1) Draw $e \sim \text{Exp}(1)$ and set the log-slice height $z = h(x_0) - e$.
- (2) Draw $u \sim \text{Unif}(0, w)$ and initialize $L = x_0 - u$, $R = x_0 + w - u$.
- (3) Step out left and right by increments of w , respecting the bounds a and b , until the interval endpoints lie below the slice or the maximum number of expansions is reached.
- (4) Truncate the interval to $[a, b]$.

- (5) Repeatedly draw $x_1 \sim \text{Unif}(L, R)$. If $h(x_1) \geq z$, accept x_1 . Otherwise shrink the interval to $[L, x_1]$ if $x_1 > x_0$ and to $[x_1, R]$ otherwise.

For dynamic and static exAL MCMC fits, this sampler is applied to the one-dimensional γ target on (L, U) by default. The same bounded-slice utility can be reused by other internal one-dimensional nonconjugate blocks; the user-facing algorithms covered here are the exAL updates described above.

7.2. Laplace-Delta approximation

For a nonconjugate block $\boldsymbol{\eta} \in \mathbb{R}^d$ with log target $\ell(\boldsymbol{\eta})$, let $\hat{\boldsymbol{\eta}}$ be a local mode and $\mathbf{H} = \nabla^2 \ell(\hat{\boldsymbol{\eta}})$. If $-\mathbf{H}$ is positive definite, the Laplace approximation is

$$q(\boldsymbol{\eta}) \approx \mathcal{N}(\hat{\boldsymbol{\eta}}, [-\mathbf{H}]^{-1}). \quad (\text{S118})$$

For a smooth function $r(\boldsymbol{\eta})$, the Delta approximation used by LDVB is

$$\mathbb{E}\{r(\boldsymbol{\eta})\} \approx r(\hat{\boldsymbol{\eta}}) + \frac{1}{2} \text{tr} [\nabla^2 r(\hat{\boldsymbol{\eta}}) [-\mathbf{H}]^{-1}]. \quad (\text{S119})$$

This is used for smooth functions of (σ, γ) in dynamic and static exAL models.

8. Package-to-algorithm map

Function	Main algorithmic role
<code>exdq1mMCMC()</code>	Dynamic DQLM/exDQLM posterior simulation: FFBS for states, Gibbs for conjugate latent/scale blocks, and bounded slice for γ when exAL is used.
<code>exdq1mLDVB()</code>	Dynamic DQLM/exDQLM mean-field VB: CAVI for Gaussian/GIG/truncated-normal blocks and Laplace-Delta for (σ, γ) in exAL.
<code>exalStaticMCMC()</code>	Static AL/exAL posterior simulation: Gaussian coefficient update, Gibbs latent/scale updates, RHS-NS inverse-gamma scale updates when selected, and bounded slice for γ in exAL.
<code>exalStaticLDVB()</code>	Static AL/exAL mean-field VB: Gaussian coefficient factor, CAVI latent and RHS-NS scale factors, and Laplace-Delta for (σ, γ) in exAL.

Table S2: Connection between the main **exdq1m** functions and the algorithmic blocks described in this supplement.

Affiliation:

Antonio Aguirre and Raquel Barata
Department of Statistics

University of California Santa Cruz
Santa Cruz, CA 95064
E-mail: jaguir26@ucsc.edu, raquel.a.barata@gmail.com